Fresh Finance

- **Private credit stocks slide after Blue Owl halts redemptions at fund**

- [US Supreme Court rules Trump's sweeping tariffs are illegal]

SHANGHAI COMPOSITE INDEX

BEM 103

Class 18: Risk Management

Hedging

Why firms hedge

Short position vs long positions: NAV=0

The drift/diffusion problem

Futures and derivatives: Execution risk

Margins as solution

Margin regulation and the financial crisis

Key concepts

- Hedging
- Long and short position
- Net Asset Position
- Margin account
- Execution risk
- Credit rating

Hedging

- The original purpose of futures contract was hedging—removing risk
  - often individuals (and firms) are forced to go short. When they are leveraged, they have potential problems of NAV=0
- Consider a firm that sells its output in advance at a fixed price
  - Profits are $\pi = Xp - C(X)$
  - You have locked in $X$and p so face uncertainty on $C(X)$
  - $C(X) = \sum p_i Q_i$
  - Hedging in this case involves using futures to get predictability on input costs.
  - Note your equity value is related to DPV of π
- Same situation if you are long on inputs but have price uncertainty on outputs.

Hedging goes both ways

- Risk on the input side, fixed price on the output
  - Alcoa and electricity prices
  - DWP and electricity prices
  - Airlines and fuel

  ….
- Fixed price on the input side, risk on the output
  - farmers in the corn belt, the wheat belt, the cotton belt…
  - Raw materials and energy producers.
  - Electronic Parts manufacturers

Hedging as risk sharing

- Start with a farmer with a silo full of wheat and a baker.
- Both face a lottery over the price of wheat next month
  - P High good for the farmer, bad for the baker
  - P Low good for the bakerm bad for the farmer
  - Now let the future price be $F^T \approx (1+r+c)^T P_0 \approx qp_h + (1-q)p_l$
- If both are risk averse then
  - $U_b(F^T) > qp_h + (1-q)p_l$
  - $U_f(F^T) > qp_h + (1-q)p_l$
- Notice the futures' market exists even if only one side is risk averse
  - If both risk neutral, then no need for futures

Hedging and returns

- When the farmer and baker enter into a futures contract, they are both perfectly hedged.
- But if you perfectly hedge your business means you remove BOTH the downside and the upside.
  - So you should be able to sell equity at a very high price
  - If your do not bear risk, they only earn the riskless rate of return
- Question do investors care?

Should firms hedge?

- Last time we looked at how an investment plan could increase their expected return by increasing leverage and how that made them more risky.
- Futures allow firms to be less risky
- Are either strategies a good idea? And if so for whom?
- Shareholders
- Principals
- Workers

Shareholders

- In principle shareholders should not care whether the firm hedges.
- If the firm's risk is diversifiable
  - Then the shareholders prefer to forego the cost of hedging.
  - Except if firm is highly leveraged and the cost of default is high.
  - But then the cost of the hedge is very high
- If the firm's risk is undiversifiable
  - Then probably no good hedge
  - And any insurance contract is just going to transfer the return to the firm that bears risk
  - So if CAPM process nothing to gain,

Principals

- If they have a large position in the firm then they are not CAPM diversified
- And they may want to hedge because they are exposed to the firm's idiosyncratic and systemic risk
- Example: farmers and other sole proprietors.
  - At issue is should they hedge the firm's position or should they hedge their position in the firm
  - It may be easier to hedge the firm than the individual's wealth
- Note workers here are like principals because its hard to diversify jobs.

Hedge funds?

- Hedge funds are not using futures in the same way as firms in the real economy
- They are trying to solve a different problem.
  - For each underlying asset there are many futures contracts (at least one per month going out 2 years)
  - Now have a bunch of arbitrage conditions.
    1. Within the set of futures of the same underlying
    2. Across futures whose underlying prices are correlated.
- Goal is to create a portfolio of futures and make money on arbitrage condition failure

Hedging and leverage

- Expand the profit function to be explicit about cost of capital.
- Pure equity firm
- $\pi(X) = Xp - C(X)$

  $\Rightarrow \pi(X)$ goes to equity holder who bear the risk
- *Leveraged firm*
- $\pi(X) = Xp - C'(X) - rD$
- Firm may want to hedge so that it has the resources to meet bond payments rD….

Futures and other risks

- Claim: The problem of risk management in portfolios is VaR or Cvar.
  - This is what the investor cares about because they are risk averse
- When there is leverage, the lender also cares about the same thing and here timing is everything.
- As we shall see the same operates for futures except its starker.

Quiz time

Detour: back to leverage decisions

- Let us assume a debt of D
  - So total long position is $V = (E_0 + D)$
  - If $D = E_0$ Leverage is 1 to 1 (debt/equity)
  - If $D = 10E_0$ Leverage is 10 to 1
- Expected return is drift-cost of capital

$$E(r) = \frac{x-y}{2}(E_0 + D) - r_i D \text{ where } i = s, l$$

- Standard term structure of interest rates => $r_s < r_l$
- Next step:
  - short term debt has higher return
  - It allows more leverage has higher return
  - It has higher risk

Optimal leverage

- Lender wants to avoid losses. Consider the problem for 1 period
- With probability 0.5 draw is down
  - $V_0 = (E_0 + D)$ $\qquad NAV_0 = E_0$
  - $V_{1d} = (E_0 + D)\,(1 - y) - Dr$
  - $NAV_{1d} = E_0 - (E_0 + D)\,y - Dr$
- Now what is D such that $NAV_1 = 0$?
  - $(E_0)(1 - y) - D(y + r) = 0$
  - $\text{First term is capital loss on equity, second term is capital loss on debt } (Dy) \text{ plus interest cost } (Dr)$
  - $D = E_0 \frac{(1-y)}{y+r}$ this the maximal debt you can take on such that $NAV_1 \geq 0$
  - $\text{That corresponds to a leverage ratio } \frac{D}{E_0} = \frac{(1-y)}{y+r}$
- Maximal leverage ratio is related to the down side only (not the return) leverage goes down as r goes up and down as the down side (y) gets worse

From Short term debt to long term

- The lender has the option not to roll over the debt at the end of every period. So the problem is 1 period.
- Fund manager maximizes return chooses leverage so that $D = E_0 \frac{(1-y)}{y+r}$
  - If down draw, then closes fund. Investors lose everything
- Choose a D such that the fund will have a positive NAV for at least three periods (or long-term debt)
  - Three consecutive down draws in a row.
  - $V_{1d} = (E_0 + D)\,(1 - y) - Dr$
  - $NAV_{1d} = V_{1d} - \mathrm{D} = E_0 - (E_0 + D)\,y - Dr$
  - $V_{t+1d} = V_t\,(1 - y) - Dr$
  - $NAV_{t+1d} = V_{t+1d} - D = V_t\,(1 - y) - D(1 + r)$

Long term debt: Expand to three periods

- $NAV_{t+1d} = V_{t+1d} - Dr = V_t\,(1-y) - D(1+r)$

- $V_{1d} = (E_0 + D)\,(1-y) - Dr = E_0\,(1-y) - D(y+r)$

- $V_{2d} = V_1(1-y) - Dr = (E_0 + D)\,(1-y)^2 - D(1+r)^2$
- $V_{2d} = E_0(1-y)^2 + D(1-y)^2 - D(1+r)^2$
- $V_{2d} = E_0(1-y)^2 + D[(1-y)^2 - (1+r)^2]$

- $V_{3d} = (E_0 + D)\,(1-y)^3 - D(1+r)^3$
- First term $(E_0 + D)\,(1-y)^3$ is gross value of fund in the worst outcome, second term $-D(1+r)^3$ is cost of capital for three period+ cost of paying back the loan.
- $NAV_{3d} > 0 \Leftrightarrow D < E_0 \frac{(1-y)^3}{(1-y)^3 - (1+r)^3}$
- $NAV_{1d} > 0 \Leftrightarrow D = E_0 \frac{(1-y)}{y+r}$
- You can check that the max D for three periods is less than for one period

Implications of the Binomial

- The lender cares only about the downside
  - With long-term debt only solution is to limit leverage
  - With short-term can call in the loan the period before a down shock brings the NAV down to zero
  - That is what lenders try to do
- Role of margins
- $D = E_0 \frac{(1-y)}{y+r} \Leftrightarrow (E_0)(1-y) = D(y+r)$
- Set the leverage so that after a down shock can pay off the loan
- Turns out this is exactly how to think about a margin

Back to the Futures

- Consider a 1-month futures contract price =100.
  - Each day the price can go up or down by a max (say 3 stdev) of y.
  - Set the margin to y.
  - if buy a 1000 notional has to put 1000y into their margin account.
  - At end of day if price went down by y, the margin account has a value of 0.
    - Choice is either to liquidate or add to margin account (another 1000y)
  - If price went up, margin account now 2000y
  - Mechanics are completely parallel to the NAV problem

Futures and margin

- Mechanics are completely parallel to the short term debt NAV problem
- Except that there is a market maker who designs the margin account.
- The goal is to keep incentives aligned so if there have been X down and 30-x up days (price is up by (30-X)y). On the last day the losing side still performs because the change is only up by one y or down by one y.

Recap

- Manage risk by managing the down side
- With binomial its just number of down draws in a row you want to survive
  - As fund manager
    - Chose debt level ratio at the beginning (and do not revise)
    - Chose a debt equity ratio and update
  - As lender can either ration credit or set appropriate premium for default risk
- Broadly the same issues come up with other distribution of risk
  - For distribution with very wide support, you can still compute probability that NAV=0